Queues and asynchronous work

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete queue-state and recovery record.

Keep job ID, payload identity, state evidence, commit, acknowledgement and separate delivery/effect counts together.

Keep one habit: Recall → W1–W3/W5 → independent three-job case → job-state project → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Caching and stale information. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Sandboxing and resource limits. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Queue**s separate acceptance from completion. Work may be redelivered, so consumers must preserve operation IDs and deduplicate committed effects.

- A **queue**d job is pending, running, completed or failed. Acknowledge only after the declared durable completion step. A repeated completed ID returns its prior result; unknown status needs reconciliation.

Caching and stale information

> **My remembered idea, example and correction**
>
> | |
> |---|
> | |
> | |
> | |
> | |

Sandboxing and resource limits

> **My remembered idea, example and correction**
>
> | |
> |---|
> | |
> | |
> | |
> | |

> **My remembered idea, example and correction**

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A queue separates accepting work from completing it. Main states are pending, running, completed and failed. Picture vocabulary queued/succeeded corresponds to pending/completed. A submission receipt proves acceptance; a worker acknowledgement is a separate signal after the declared durable completion step. Main job J commits a checked result then crashes before acknowledgement.

  Redelivery with the same ID and payload consults reliable completion evidence, returns the prior result and acknowledges without another effect: two deliveries, one mock effect. Unknown or running-only status requires reconciliation; same-ID changed payload is a conflict. Separate J12 receipt proves queued only; checked output permits succeeded.

  An observed pre-output crash permits failed status, while an unobserved crash may leave stale running status. No retry policy is implied. Reliable effect/status correspondence is an explicit assumption. The code is sequential in-memory fiction with copied snapshots; it implements no broker, durable commit, concurrency or exactly-once delivery. Losing history can lead to a repeated mock effect.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A queue separates accepting work from completing it. Main states are pending, running, completed and failed. Picture vocabulary queued/succeeded corresponds to pending/completed. A submission receipt proves acceptance; a worker acknowledgement is a separate signal after the declared durable completion step. Main job J commits a checked result then crashes before acknowledgement.

  Redelivery with the same ID and payload consults reliable completion evidence, returns the prior result and acknowledges without another effect: two deliveries, one mock effect. Unknown or running-only status requires reconciliation; same-ID changed payload is a conflict. Separate J12 receipt proves queued only; checked output permits succeeded.

  An observed pre-output crash permits failed status, while an unobserved crash may leave stale running status. No retry policy is implied. Reliable effect/status correspondence is an explicit assumption. The code is sequential in-memory fiction with copied snapshots; it implements no broker, durable commit, concurrency or exactly-once delivery. Losing history can lead to a repeated mock effect.

- Worker commits job J then crashes before **queue acknowledgement**. J is delivered again. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A queue separates accepting work from completing it. Main states are pending, running, completed and failed. Picture vocabulary queued/succeeded corresponds to pending/completed. A submission receipt proves acceptance; a worker acknowledgement is a separate signal after the declared durable completion step. Main job J commits a checked result then crashes before acknowledgement.

  Redelivery with the same ID and payload consults reliable completion evidence, returns the prior result and acknowledges without another effect: two deliveries, one mock effect. Unknown or running-only status requires reconciliation; same-ID changed payload is a conflict. Separate J12 receipt proves queued only; checked output permits succeeded.

  An observed pre-output crash permits failed status, while an unobserved crash may leave stale running status. No retry policy is implied. Reliable effect/status correspondence is an explicit assumption. The code is sequential in-memory fiction with copied snapshots; it implements no broker, durable commit, concurrency or exactly-once delivery. Losing history can lead to a repeated mock effect.

- A report omits this qualification: **Queue** ordering and exactly-once properties must be declared, not assumed from a simple mock. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A queue separates accepting work from completing it. Main states are pending, running, completed and failed. Picture vocabulary queued/succeeded corresponds to pending/completed. A submission receipt proves acceptance; a worker acknowledgement is a separate signal after the declared durable completion step. Main job J commits a checked result then crashes before acknowledgement.

  Redelivery with the same ID and payload consults reliable completion evidence, returns the prior result and acknowledges without another effect: two deliveries, one mock effect. Unknown or running-only status requires reconciliation; same-ID changed payload is a conflict. Separate J12 receipt proves queued only; checked output permits succeeded.

  An observed pre-output crash permits failed status, while an unobserved crash may leave stale running status. No retry policy is implied. Reliable effect/status correspondence is an explicit assumption. The code is sequential in-memory fiction with copied snapshots; it implements no broker, durable commit, concurrency or exactly-once delivery. Losing history can lead to a repeated mock effect.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may success be claimed for all accepted?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **queue**s and asynchronous work **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **queue**s and asynchronous work.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Queue**s and asynchronous work. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Queue**s and asynchronous work. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Jobs A, B, C accepted; A completed, B failed, C pending. How many completed, and may success be claimed for all accepted?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

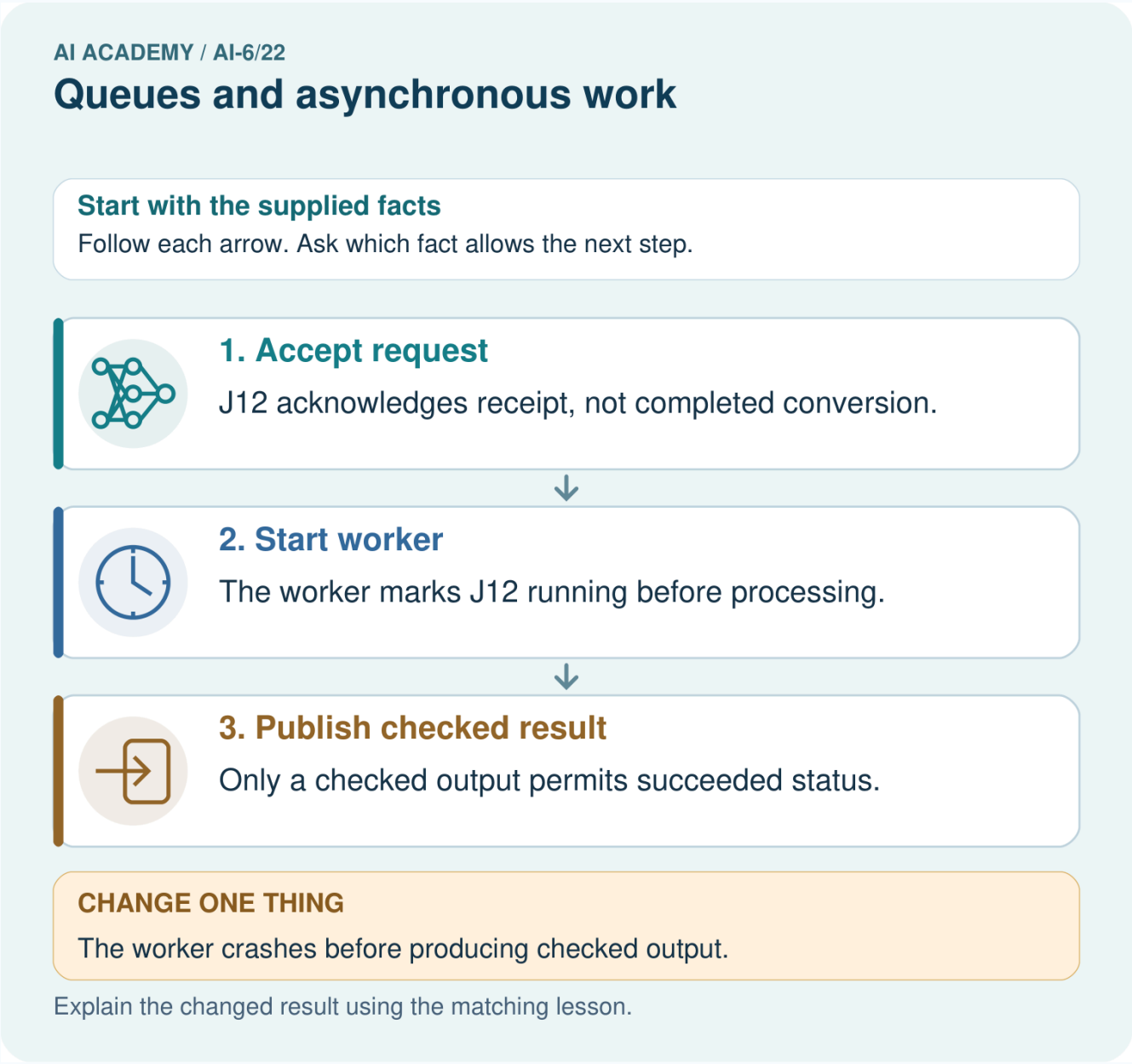
Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- **Queue**d is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission.
- States: **queue**d, running, succeeded, failed. Submission returns J12 with **queue**d status.
- Explain what the **queue**d J12 submission response proves and what it does not. Which event permits succeeded status, and what status is honest after a worker crash?
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- **Queue**d is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission.
- Which labelled step matches this detail? Only a checked **output** permits succeeded status.
- 1. Start worker
- 2. Publish checked result
- 3. Accept request

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Publish checked result Only a checked **output** permits succeeded status. Asynchronous work separates submission from completion.

Honest status and a job identifier let users track that difference.

Changed-case question and optional hint

- Changed-case practice: The worker crashes before producing checked **output**. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. The job is failed or awaiting a controlled retry, not succeeded.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The job is failed or awaiting a controlled retry, not succeeded. Receipt is not **evidence** that conversion finished.

Repair and retain the record

Review Caching and stale information (ai-6/21). Goal: Explain the **mechanism** and terminology of caching and stale information. Short practice: Identify the version mismatch in the cached 09:00 answer.

What should the first version -5 request return, and what permits a later version -5 cache hit?. Return to this week and explain what you changed.

Review Sandboxing and resource limits (ai-6/19). Goal: Explain the **mechanism** and terminology of sandboxing and resource limits. Short practice: Check A against both resource limits and B against network isolation.

Why must A stop after three seconds even if its **output** remains only 2 MB?. Return to this week and explain what you changed.

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

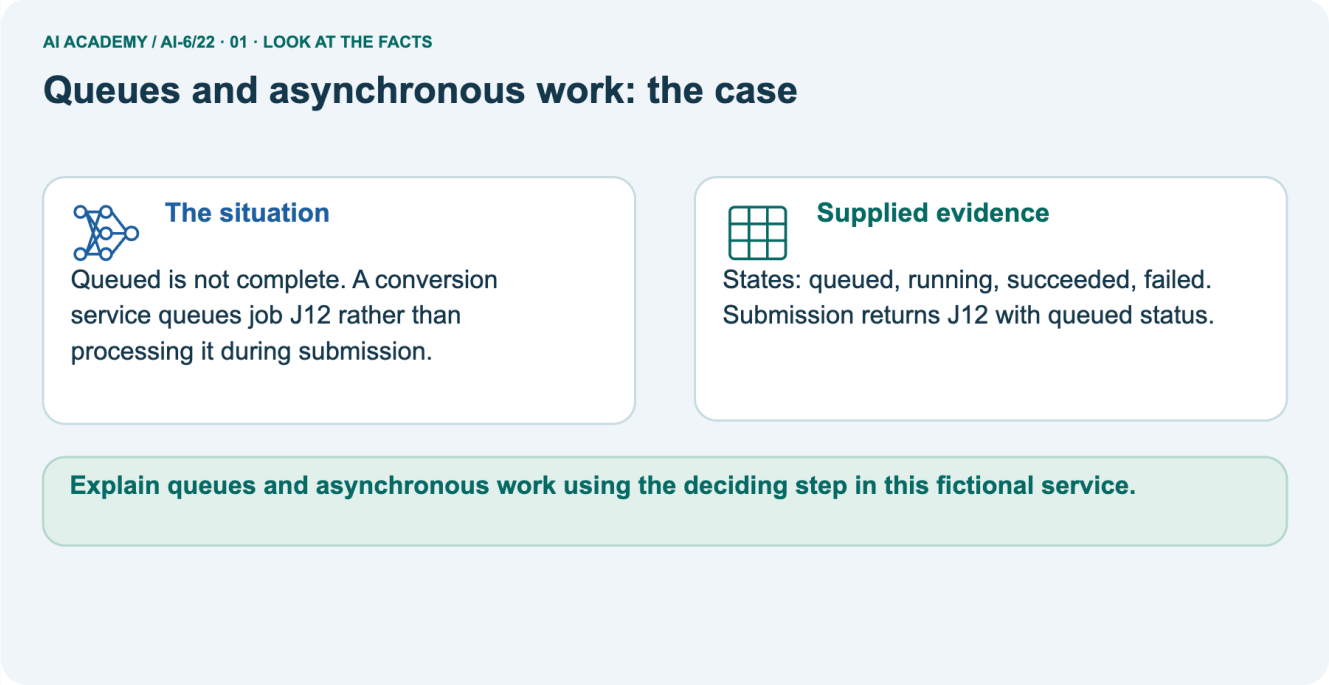
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Queues and asynchronous work: the case



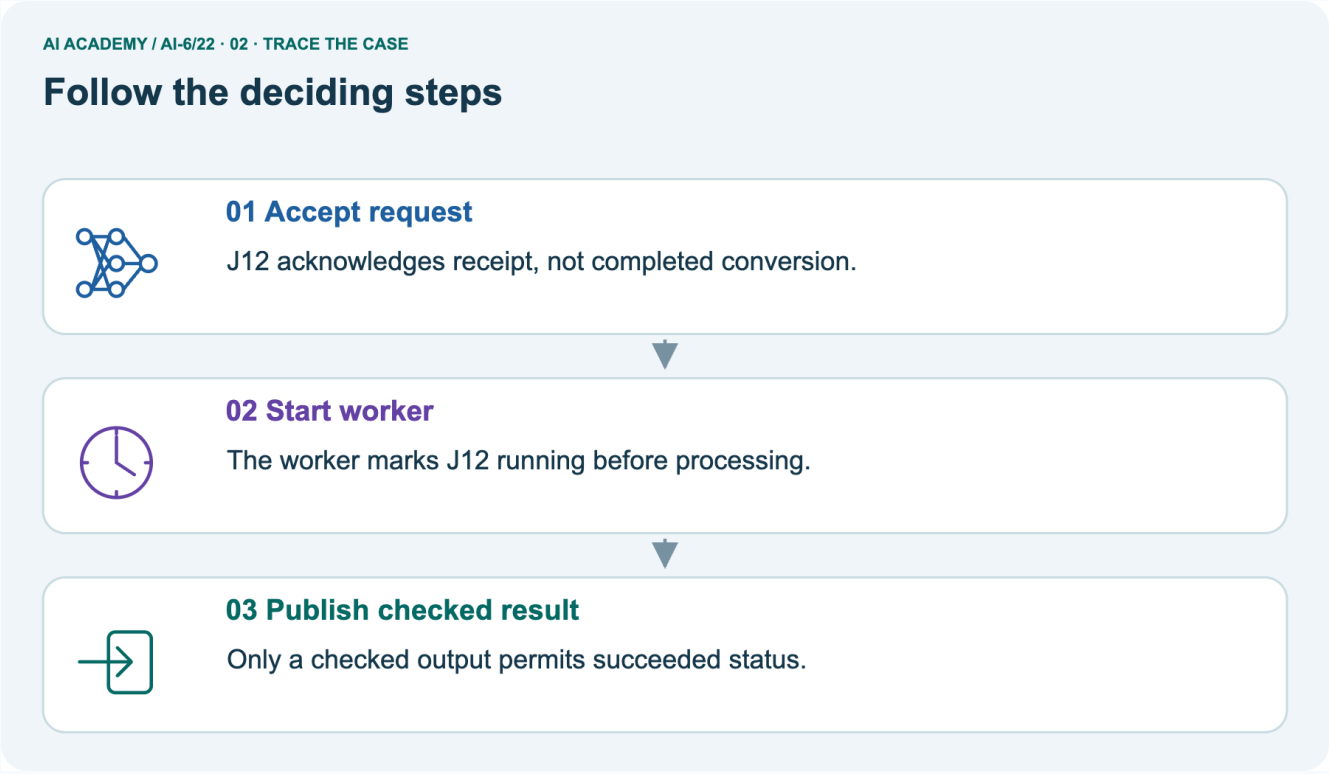
Follow the labelled picture with the matching explanation. Open the vector diagram

Queued is not complete. A conversion service **queue**s job J12 rather than **process**ing it during submission. Supplied facts: States: **queue**d, running, succeeded, failed. Submission returns J12 with **queue**d status.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



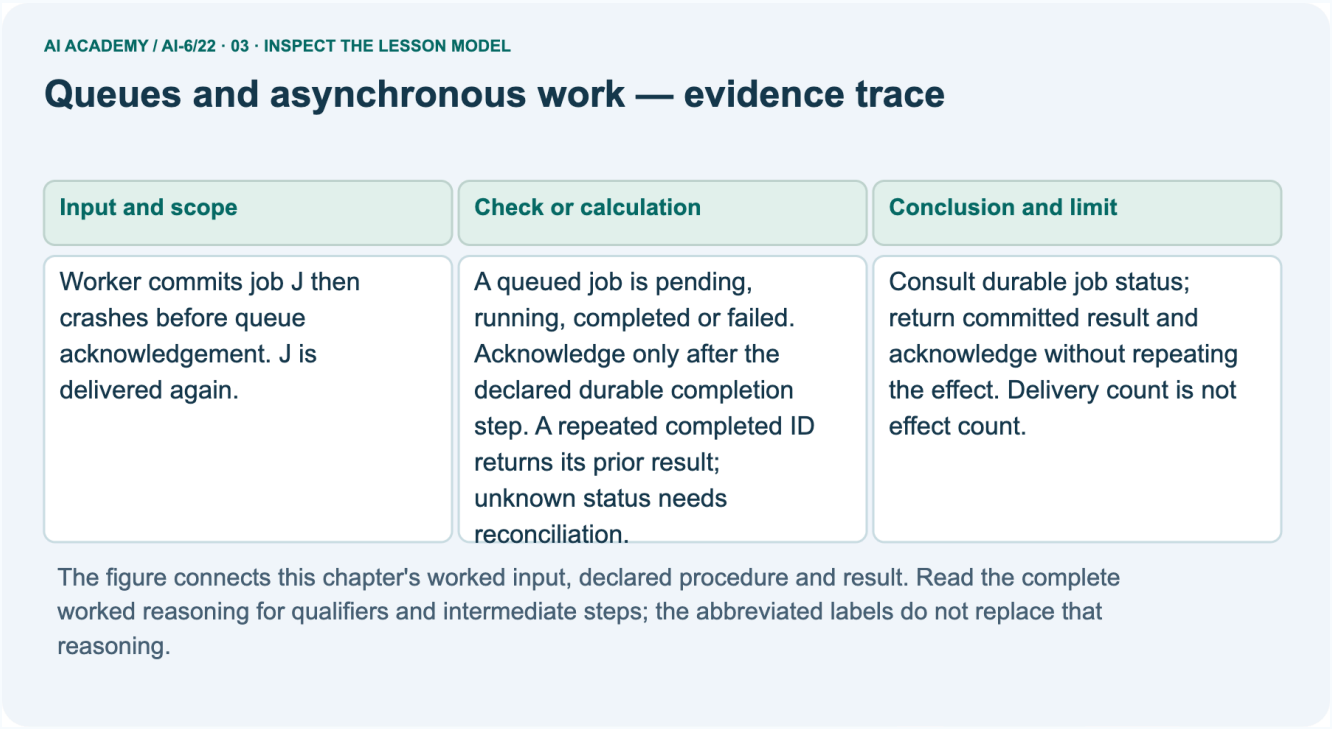
Follow the labelled picture with the matching explanation. Open the vector diagram

Accept request: J12 acknowledges receipt, not completed conversion. → Start worker: The worker marks J12 running before **process**ing. → Publish checked result: Only a checked **output** permits succeeded status.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Queues and asynchronous work — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Queues and asynchronous work — **evidence** trace. Worker commits job J then crashes before **queue acknowledgement**. J is delivered again. | A **queue**d job is pending, running, completed or failed.

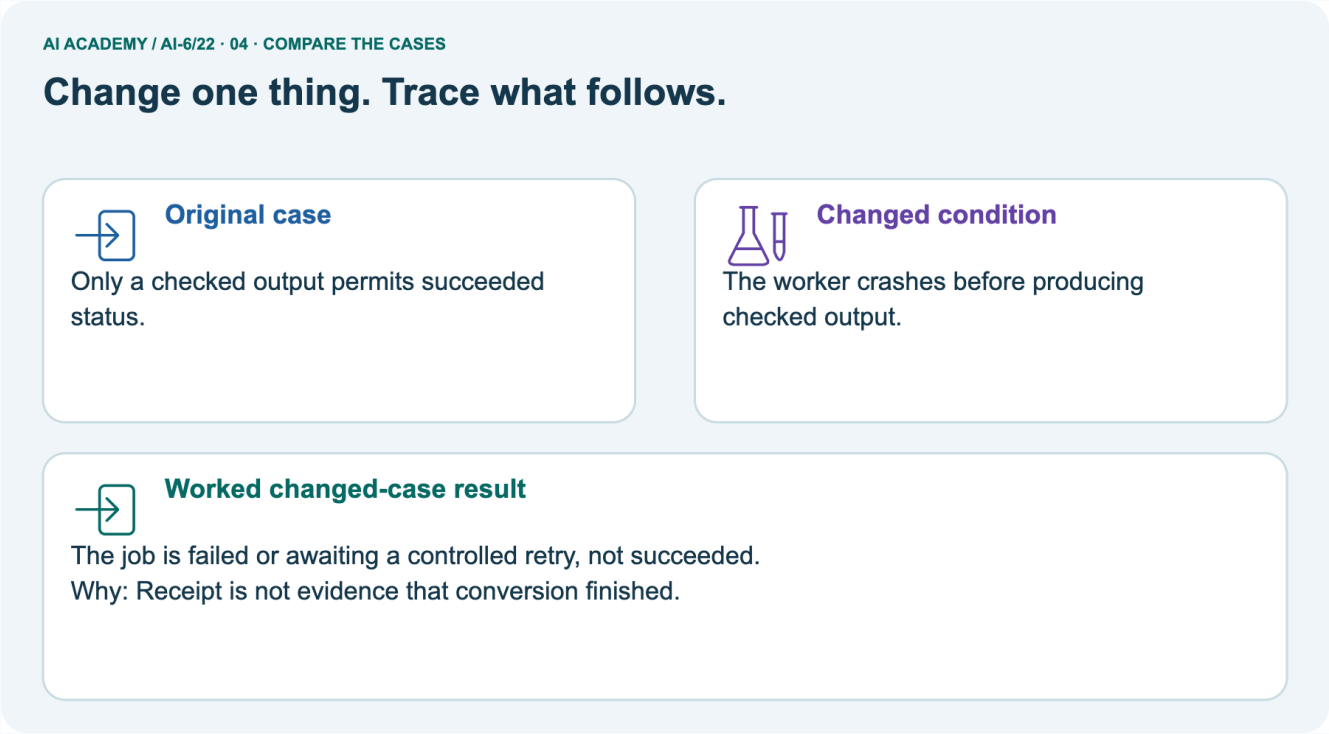
Acknowledge only after the declared durable completion step.

A repeated completed ID returns its prior result; unknown status needs reconciliation. | Consult durable job status; return committed result and acknowledge without repeating the effect. Delivery count is not effect count.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

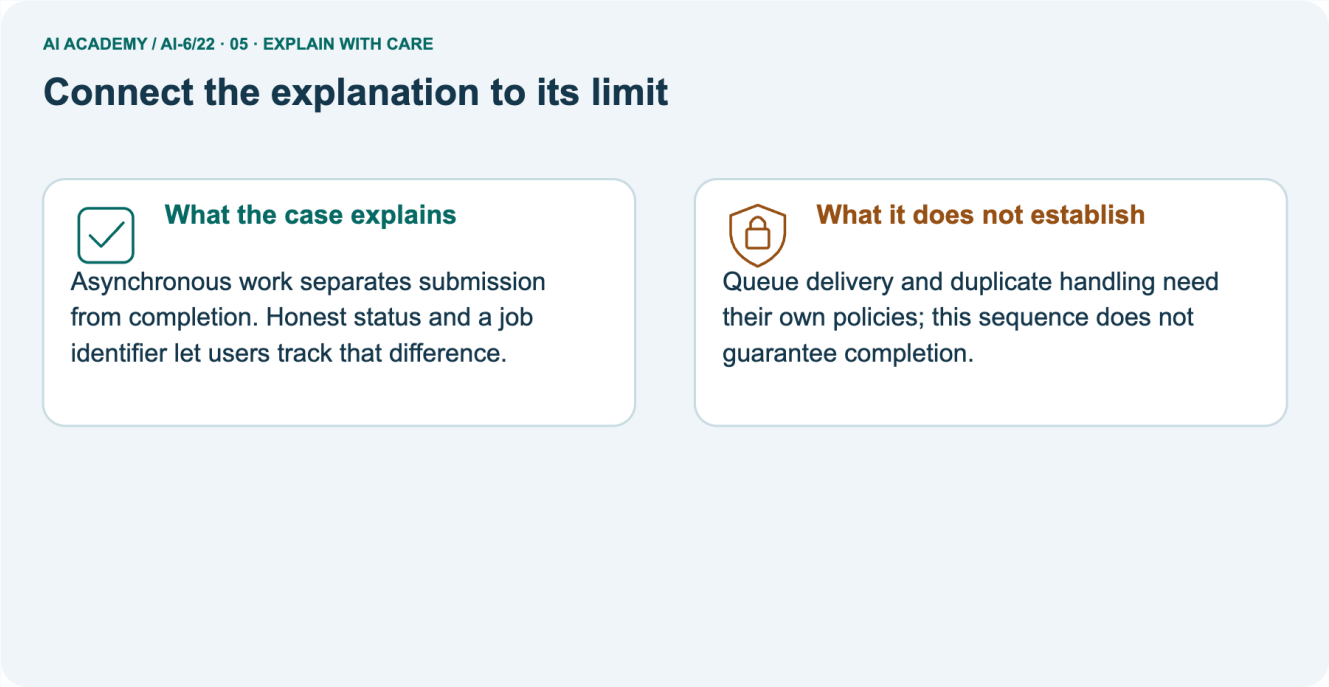
Original case: Only a checked **output** permits succeeded status. Changed condition: The worker crashes before producing checked **output**. Expected result: The job is failed or awaiting a controlled retry, not succeeded.

Reason: Receipt is not **evidence** that conversion finished.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

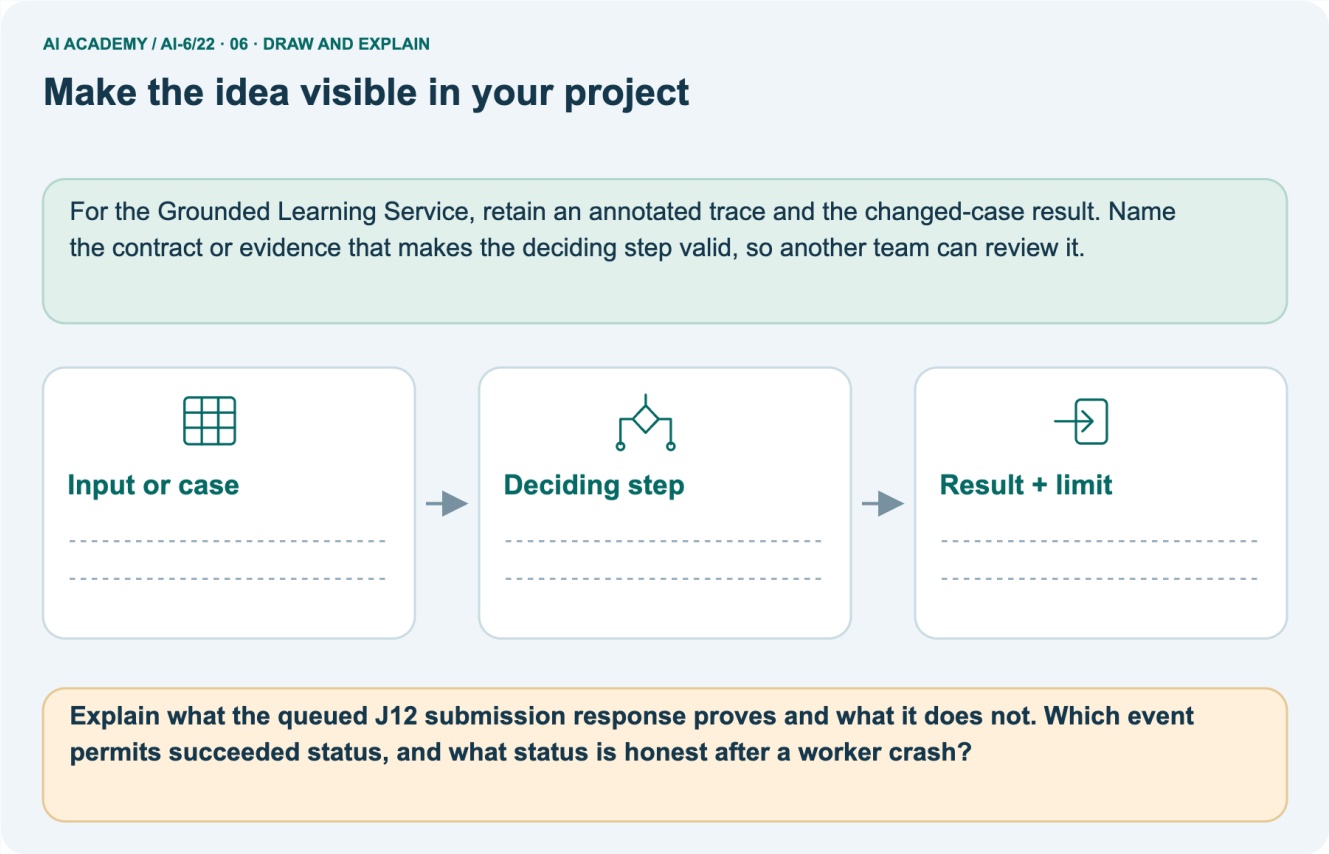
Asynchronous work separates submission from completion. Honest status and a job identifier let users track that difference.

Limit: **Queue** delivery and duplicate handling need their own policies; this sequence does not guarantee completion.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Explain what the **queue**d J12 submission response proves and what it does not.

Which event permits succeeded status, and what status is honest after a worker crash?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Queues and asynchronous work: the case

02. Follow the deciding steps

03. Queues and asynchronous work — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.